InHand Networks

Flexible Deployment, Reliable Connectivity

ODU302 4G Outdoor Router

- 4G LTE
- Wi-Fi
- IP65 Rated
- Cloud Managed

1. Product Overview

The ODU302 is an outdoor cellular router designed to simplify outdoor networking deployment. Conventional outdoor connectivity solutions require combining a standard router, external antennas, feeder cables, and a weatherproof enclosure—involving a wide variety of accessories, multiple cabling and installation steps, and correspondingly high complexity for troubleshooting and maintenance. The ODU302 integrates 4G cellular, built-in omnidirectional antennas, and an IP65-rated enclosure into a single device, replacing the traditional multi-component setup and thereby eliminating peripheral accessories and protective hardware at the source. The device supports top-mount, wall-mount, and pole-mount installation, and can be mounted directly on cabinet tops, exterior walls, or surveillance poles—removing the need for in-cabinet space and feed-through drilling. Equipped with multiple failover and self-healing mechanisms including link detection, auto-redial, dual-SIM switching, and a hardware watchdog, the device automatically restores connectivity upon network interruption. Coupled with the Device Manager cloud management platform, it enables centralized remote operation and maintenance across multi-site, large-scale deployments. From on-site deployment to remote O&M, the ODU302 makes outdoor connectivity simpler.

Key Features

- **High-Speed 4G Access**: Supports 4G LTE cellular access and delivers omnidirectional signal coverage via built-in antennas, greatly simplifying deployment while ensuring a stable network connection.
- **Wi-Fi**: Supports 802.11b/g/n, providing stable and reliable wireless connectivity.
- **Always-On Reliability**: Link backup, dual-SIM failover, heartbeat detection, and a hardware watchdog together guarantee communication continuity.
- **Security and Trust**: Enterprise-grade firewall and a full suite of mainstream VPN tunneling protocols meet the compliance requirements for encrypted business data transmission.
- **Environmental Robustness**: IP65 all-weather protection with a wide operating temperature range of -40 °C to +70 °C, withstanding harsh outdoor conditions

such as direct sunlight, extreme cold, dust, and rain.
- **Cloud-Based O&M**: Integrates with the Device Manager cloud platform for batch remote management and maintenance.
- **Flexible Installation for All Scenarios**: Compact and lightweight form factor supports wall-mount, pole-mount, and top-mount installation, adapting to a full range of deployment scenarios.
- **Dual Redundant Power Inputs**: Compatible with both IEEE 802.3af standard PoE power over Ethernet and a wide-range DC 9–36 V terminal power input.

Typical Application Scenarios

Small Campuses and Public Amenities

Small campuses and outdoor public-service facilities are typically characterized by a large number of distributed sites, where full-coverage wired deployment is costly and centralized remote management is required. The ODU302 supports wall-mount, pole-mount, and top-mount installation, flexibly adapting to exterior walls, poles, and other deployment scenarios; its multi-layer link redundancy ensures 7×24 continuous service availability. Compatible with the InHand cloud management platform, it enables centralized monitoring of thousands of endpoints, batch configuration delivery, and unified firmware push. Combined with VPN tunneling for remote private-network interconnection, it fully satisfies the requirements of large-scale centralized O&M.

Warehousing and Logistics Parks

Steel-structured plant walls tend to block cellular signals, limiting indoor router throughput, while on-site scanning terminals and surveillance equipment require a mixed wired/wireless network. The device can be installed on exterior walls, rooftops, or park poles, leveraging its built-in omnidirectional antenna to access the 4G network and providing stable connectivity to local equipment via Ethernet ports and Wi-Fi.

Outdoor Equipment Connectivity

Outdoor terminals such as ATM cabinets and fuel dispensers have limited internal space. Traditional routers with external antennas not only occupy scarce cabinet space but also require additional drilling and cabling. The ODU302 supports top-mount deployment, integrates omnidirectional cellular antennas, and features IP65 dust and water protection, allowing it to be mounted externally on the terminal enclosure—reducing cabinet drilling and internal wiring while simplifying deployment cost.

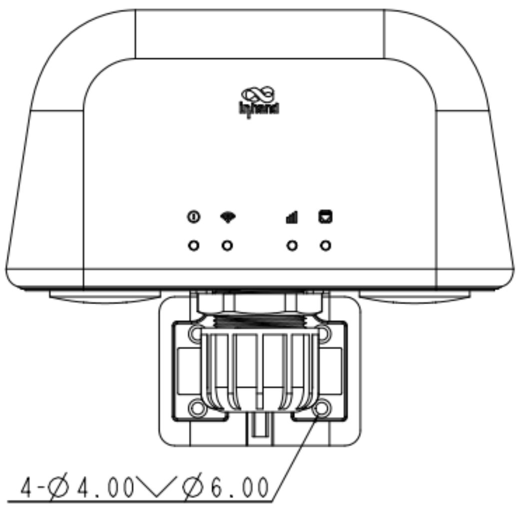
Outdoor Security and Surveillance

Outdoor surveillance sites are typically deployed on open poles and largely lack wired broadband resources. In conventional setups, routers are placed inside separate weatherproof boxes whose enclosures tend to shield cellular signals and degrade network quality. The ODU302 is inherently IP65 dust- and water-resistant and operates reliably from -40 °C to +70 °C, requiring no additional weatherproof enclosure and supporting direct outdoor exposure. Externally mounted with built-in omnidirectional antennas, it achieves stronger cellular reception and ensures stable upload of surveillance video streams.

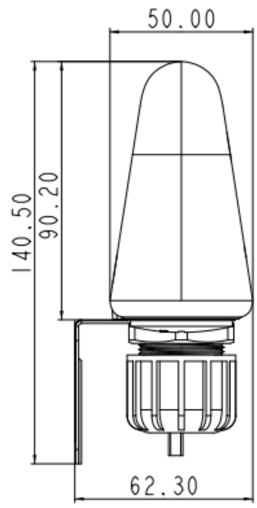
Signal-Constrained Sites Such as Underground Parking

Concrete walls and metal structures in underground parking facilities readily shield cellular signals, and wired broadband is generally unavailable. Meanwhile, license-plate recognition, surveillance, and barrier-gate equipment are numerous, making cabling difficult and inspection costly. The ODU302 can be deployed at signal-reachable areas such as garage entrances/exits and ramps, receiving cellular signals across the area via its built-in omnidirectional antenna to ensure stable online operation for parking management and video backhaul.

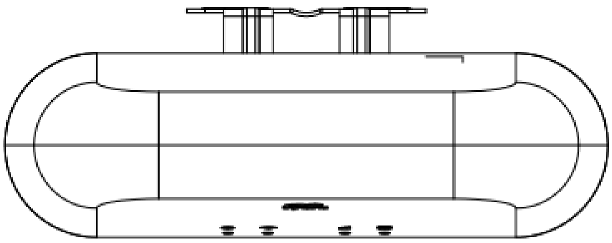
Product Dimensions

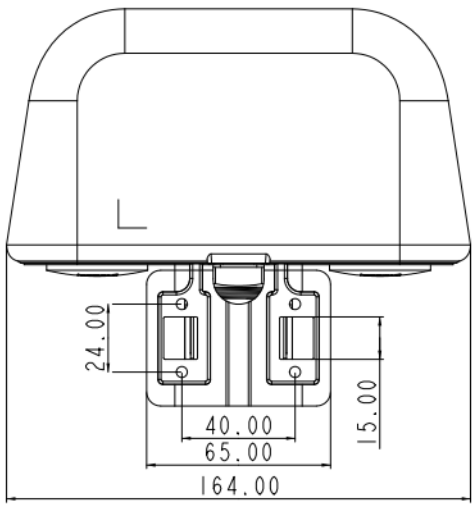


Front View



Side View


Bottom View



Rear View

Note:

1. All dimensions are in millimeters (mm).
2. Dimensions (L × W × H): 164 × 50 × 129.7 mm.
3. All dimensions are approximate and **for reference only**.
4. The dimensional drawings **shall not be used for production machining**.
5. Dimensions are subject to change without prior notice.

2. Product Highlights

Highlight 1: One Device, a Complete Outdoor Connectivity Solution

Built on an all-weather outdoor operation foundation, the ODU302 integrates 4G cellular, built-in omnidirectional antennas, and an IP65-rated enclosure—eliminating the need for external antennas, feeder cables, and weatherproof boxes, and reducing peripheral accessories and on-site protective components at the source.

Highlight 2: Three Installation Methods for Full-Scenario Adaptability

With a compact and lightweight form factor, the device supports top-mount, wall-mount, and pole-mount installation. It can be mounted directly on cabinet tops, exterior walls, or surveillance poles, removing the need for in-cabinet space and feed-through drilling, and providing a simple installation solution for every outdoor connectivity scenario.

Highlight 3: Enterprise-Grade Security and Scalable Cloud O&M

The device integrates an enterprise-grade stateful inspection firewall and a full suite of mainstream VPN tunneling protocols, effectively safeguarding network and data transmission security.

Highlight 4: Multi-Layer Link Redundancy for Multiple Uptime Assurances

The device supports a three-tier protection logic: hardware watchdog self-check and recovery, dual-SIM cellular backup, and cellular/wired dual-link mutual backup. SIM signal degradation, dial-up failure, or wired broadband interruption all trigger seamless switching to the backup channel, ensuring uninterrupted service. This is well suited to high-availability scenarios such as transaction terminals, warehousing equipment, and outdoor surveillance.

Highlight 5: Cloud-Based Batch Management and O&M

The ODU302 natively supports the InHand Device Manager cloud platform, enabling centralized management of distributed devices at scale—remote configuration, firmware upgrade, traffic statistics, and fault alarm—significantly reducing the manual effort of multi-site field O&M.

Highlight 6: Dual Redundant Power for Single-Point Power-Failure Protection

The device is compatible with both IEEE 802.3af standard PoE power over Ethernet and a wide-range DC 9–36 V terminal input. The two power sources are monitored in parallel in real time, with millisecond-level automatic switchover upon failure of either source. This adapts to various on-site power schemes such as shop PoE tapping, equipment weak-current cabinets, and solar power, ideally fitting unattended long-term online scenarios.

3. Hardware Specifications

3.1 Hardware Overview

Parameter	Specification
Processor	580 MHz
Memory	128 MB DDR2 SDRAM

Parameter	Specification
Storage	32 MB SPI NOR Flash
4G Module	4G LTE Cat.4, 150 Mbps DL / 50 Mbps UL
Recommended User Count	32
Ethernet Ports	2 × 10/100 Mbps RJ45, WAN/LAN switchable, LAN2 supports PoE input
SIM Slots	2 × Nano SIM (drawer-type socket), dual-card failover
Digital I/O	2 × configurable as DI or DO
Cellular Antenna	2 × built-in 4G antennas (ANT_4G_1, ANT_4G_2)
Wi-Fi Antenna	1 × built-in Wi-Fi antenna
Wi-Fi	IEEE 802.11 b/g/n, 2.4 GHz, up to 150 Mbps
Wi-Fi Transmit Power	802.11b: 16 dBm ± 2 dBm (11 Mbps) 802.11g: 16 dBm ± 2 dBm (54 Mbps) 802.11n@2.4 GHz HT20 MCS7: 16 dBm ± 2 dBm 802.11n@2.4 GHz HT40 MCS7: 16 dBm ± 2 dBm
LED Indicators	System, Wi-Fi, Cellular, Ethernet
Reset Button	Pin-hole Reset button
Power Input	DC 9–36 V (4-pin combined power and I/O terminal, over-current and reverse-polarity protected) IEEE 802.3af PoE input
Standby Power Consumption	90–110 mA @ 12 V
Operating Power Consumption	280–300 mA @ 12 V

Parameter	Specification
Peak Power Consumption	340–360 mA @ 12 V
Ingress Protection	IP65
Enclosure	Plastic
Operating Temperature	-40 °C ~ +70 °C
Storage Temperature	-40 °C ~ +85 °C
Humidity	5% ~ 95% (non-condensing)
Dimensions	164 × 50 × 129.7 mm
Weight	Approx. 500 g
Installation	Wall-mount, pole-mount, top-mount

4. Network Connectivity

4.1 Cellular Network

- **Network Modes:** GSM/GPRS/EDGE, UMTS/HSPA+/EVDO, TD-SCDMA, FDD LTE/TDD LTE
- **Network Access:** APN, VPDN
- **Access Authentication:** CHAP/PAP
- **Connection Modes:** Always-on, dial-on-demand, manual dial-up
- **Activation Methods:** Data-triggered, SMS-triggered

4.2 SIM Card Management

- **Dual-SIM Support:** Drawer-type dual Nano SIM slots
- **Dual-SIM Failover:** Primary/backup SIM automatic switching supported
- **Intelligent Switching:**
  - Automatic SIM switch upon dial-up failure
  - Automatic switch when signal falls below threshold
  - Configurable primary/backup SIM and maximum dial-up attempts
- **SIM Binding:** ICCID binding, PIN code management

4.3 Wired Network

- **WAN Protocols:** Static IP, DHCP, PPPoE
- **LAN Protocols:** ARP, Ethernet
- **Interface Mode:** 2 × RJ45 ports, WAN/LAN configurable
- **NAT:** Network Address Translation (NAPT)

4.4 Wi-Fi 4 Wireless Access

- **Standard:** IEEE 802.11 b/g/n (Wi-Fi 4)
- **Frequency Band:** 2.4 GHz single band
- **Max Rate:** 150 Mbps
- **Modes:** Access Point (AP), Station (STA)

- **Security:** Open system, shared key, WPA/WPA2
- **Encryption:** WEP/TKIP/AES

5. Link Management and Backup

5.1 Intelligent Link Backup

- **Backup Mode:** Cellular dial-up and WAN port can serve as primary/backup for each other
- **Detection Methods:** ICMP/TCP/DNS link-quality detection (configurable)
- **Switching Policy:** Automatic link-failure switchover, auto-redial on disconnection

5.2 VRRP Hot Standby

- **Protocol Support:** VRRP (Virtual Router Redundancy Protocol)
- **Application Scenario:** Dual-device hot standby for improved service continuity

5.3 Link Quality Assurance

- **Link Detection:** Heartbeat packet monitoring with auto-reconnect on disconnection
- **Dual-SIM Switching:** Automatic switch to backup SIM upon cellular link or dial-up anomaly
- **Built-in Watchdog:** Device self-check with automatic fault recovery

6. Security Features

6.1 Firewall

- **Stateful Inspection:** SPI full stateful packet inspection
- **DoS Protection:** Denial-of-service attack defense
- **Access Control:** ACL access control list

- **Content Filtering:** URL content filtering
- **Port Mapping:** Port forwarding, virtual IP mapping
- **IP-MAC Binding:** IP-MAC binding; multicast/Ping packet filtering
- **NAT:** NAPT supported

6.2 VPN Support

- IPsec VPN (IKEv1, IKEv2)
- L2TP VPN (client)
- PPTP VPN (client)
- GRE tunnel
- OpenVPN (client/server)
- WireGuard
- ZeroTier
- CA digital certificate support

6.3 Traffic Shaping (QoS)

- **Bandwidth Control:** Interface-level bandwidth limiting
- **Per-IP Rate Limiting:** IP-based uplink/downlink rate limiting
- **QoS Policy:** Bandwidth and rate limiting policies

7. Local Network Services

7.1 DHCP Service

- **DHCP Server:** Dynamic IP address assignment
- **DHCP Relay:** Cross-subnet DHCP relay
- **DHCP Client:** Client address acquisition

7.2 DNS Service

- **DNS Relay/Proxy:** Local caching and forwarding
- **Custom DNS:** Configurable DNS servers

7.3 Dynamic DNS (DDNS)

- **Multi-Provider Support:** Mainstream DDNS service providers
- **Automatic Update:** Automatic DNS record update upon WAN/cellular IP change

7.4 Static Routing

- **Routing Entries:** Static route configuration
- **Default Route:** Cellular/WAN default route supported

7.5 IP Passthrough

- **Bridge Mode:** WAN/dial-up IP passthrough to downstream devices
- **Flexible Configuration:** Passthrough mode available on both dial-up and WAN ports

8. Cloud Management and O&M

The ODU302 integrates with the InHand Device Manager cloud management platform to deliver unified remote O&M for large-scale distributed outdoor devices.

8.1 Device Manager Cloud Management

Feature	Description
Device Onboarding	Device Manager cloud platform and custom server support
Remote Management	Device status query, remote configuration delivery, firmware upgrade, remote reboot
Batch O&M	Batch configuration and maintenance for large-scale device fleets

Feature	Description
SNMP Management	SNMP v1/v2c/v3, with Trap alarm reporting
Traffic Management	Traffic threshold, traffic statistics, and traffic alarm
Alarm Types	System reboot, LAN port up/down, traffic alarm, SIM card failure
Notification Method	Email alarm (configurable)

9. System Management and Maintenance

9.1 Device Management

- **Web Management:** Graphical Web configuration interface
- **CLI Management:** Telnet, SSH, Console
- **Configuration Import/Export:** Backup and restore
- **Factory Reset:** Pin-hole reset or Web factory reset
- **Firmware Upgrade:** Web or Device Manager remote upgrade

9.2 Logging and Diagnostics

- **System Log:** Local syslog, remote log, and serial-port log export; key logs persisted across power loss
- **Diagnostic Tools:** Ping (connectivity), Traceroute (route tracing), network speed test (link performance)

9.3 SMS Functions

- **Status Query:** Query device status via SMS
- **Remote Reboot:** Trigger device reboot via SMS

9.4 Other System Capabilities

- **Dial-on-Demand:** Dial-on-demand; data/SMS activation
- **Status Query:** System status, Modem status, network connection status, routing status
- **Scheduled Maintenance:** NTP time synchronization; automatic fault recovery (built-in watchdog)

10. Ordering Information

10.1 Product Models

Model	Region	Bands
ODU302-CNC4	China	LTE-FDD: B1/B3/B5/B8 LTE-TDD: B34/B38/B39/B40/B41 WCDMA: B1/B8 GSM: B3/B8 CDMA: BC0 TD-SCDMA: B34/B39
ODU302-NAC4	North America	LTE-FDD: B2/B4/B5/B12/B13/B14/B66/B71 WCDMA: B2/B4/B5
ODU302-EUC4	Europe/Asia-Pacific	LTE-FDD: B1/B3/B5/B7/B8/B20/B28 LTE-TDD: B38/B40/B41 WCDMA: B1/B5/B8 GSM/EDGE: B3/B8

10.2 Package Contents

- ODU302 main unit × 1
- IEEE 802.3af PoE adapter × 1
- 1 m Ethernet cable × 1
- Mounting kit (wall-mount/pole-mount/top-mount) × 1

- Quick Start Guide × 1

11. Reliability Standards and Certifications

11.1 Ingress Protection

- **IP Rating:** IP65 (dust-tight, protected against high-pressure water jets)

11.2 Mechanical Reliability

Test Item	Standard
Shock	IEC 60068-2-27
Vibration	IEC 60068-2-6
Free Fall	IEC 60068-2-32

11.3 EMC Immunity

Test Item	Standard
ESD (Electrostatic Discharge)	EN 61000-4-2, Level 2
Radiated RF Immunity	EN 61000-4-3, Level 2
EFT/Burst	EN 61000-4-4, Level 2
Surge	EN 61000-4-5, Level 2
Conducted RF Immunity	EN 61000-4-6, Level 2
Oscillatory Wave Immunity	EN 61000-4-12, Level 2
Power-Frequency Magnetic Field Immunity	EN 61000-4-8, 400 A/m

11.4 Certifications and Compliance

Certification	Status
FCC	Completed
IC	Completed
PTCRB	Completed
AT&T	In-progress
Verizon	In-progress
T-Mobile	In-progress
CE	In-progress

> Note: Certification status is subject to the time of publication. For the latest certification status of specific SKUs, please contact the InHand sales team.

12. Contact Us

InHand Networks

- **Website:** www.inhand.com
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